

SANKUR KUNDU

Machine Learning Engineer · Software Engineer · Data Scientist

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Engineer who ships systems end to end. Co-founder and Director of Technology at Gigzen, where I built both products, the Postgres they share, and the public record of every test run against them — including the ones that found defects in my own work. In public use: on npm, as a Windows app, and a signed Android build.

EXPERIENCE

Gigzen Private Limited · Co-founder & Director, Technology

Bhubaneswar | Apr 2026 – Present

- Co-founded an MCA-registered company targeting the **USD 582B gig economy**; lead the engineering from schema to signed Android artefact.
- Run the Postgres database both products share — **20 tables under 49 row-level-security policies** — verified by attack: one authenticated driver reads 0 of another's messages and is refused (403) on writes.
- Sustained **200 concurrent simulated drivers against a local backend across 932,455 API calls with zero failures and p95 latency under 900 ms**, all 13 methods, sessions refreshed mid-run, every account created and removed.
- Publish what failed: the engineering record (shakhtar-sankur.github.io/gigzen/test-report.html) names withdrawn claims and marks unreproducible numbers as not measured.
- Shipped Populace Studio, an Electron app over the same engine — live per-endpoint telemetry, a real-time world map of the simulated population, and reporting that survives a crashed run; eleven releases, each with its SHA-256.
- Load-tested over a documented day of seven runs — 3,174,576 API calls, zero failures, every run published including the two rejected — and measured the limits of scalability: 178, 241 then 272 calls/sec at 100, 200 and 300 drivers; saturation, not collapse.

Freelance Data Scientist · Contract via Upwork

Remote, client in Spain | Jun 2026 – Present

- Engineer Python/Pandas/NumPy/SQL pipelines over multi-GB census, commercial and STATA .dta sources, turning unstructured records into analysis-ready datasets.
- Build validation and reproducibility checks cross-functionally, collaborating with researchers and statisticians so results can be re-derived, not just reported.
- Deliver statistical modeling inputs and analytical outputs to an international research team under fixed publication timelines.

PRODUCTS

Populace · Multi-user testing platform

run it: `npx @gigzen/populace demo` · npm: `@gigzen/populace` · [Shakhtar-Sankur/populace](https://shakhtar-sankur.github.io/populace)

- Simulated populations that drive a real app through its own API as authenticated users. **101 self-tests**, zero runtime dependencies, CI on Node 18 and 22, published open source under AGPL-3.0.
- App-agnostic engine behind a **13-method adapter contract**; proven portable against Gitea, an unrelated Go app; failures come with a cause and a fix, not an error string. Found 6 real defects in a finished, manually tested app — five in the first three and a half minutes.
- Per-endpoint p50/p95/p99 latency instrumentation and three independent guards against production, two refusing outright; empty adapter methods are not coverage: a fresh scaffold reports 2/13 and refuses to run.

Waggle · Gig-worker platform, Android

install it: shakhtar-sankur.github.io/waggle · [Shakhtar-Sankur/waggle](https://shakhtar-sankur.github.io/waggle)

- React + TypeScript + Capacitor on Supabase: realtime feed, group chat with read receipts and voice notes, GPS tracking map-matched to roads, speed-classified and browsable by day, an earnings heatmap and turn-by-turn directions. Offline writes queue and send on reconnect; 8.2 MB signed build.
- Localised into **43 languages, 8 of them at 93% or better** against a 391-key base including consent and legal text, mirrored right-to-left for Arabic, Urdu and Hebrew, currency and platforms resolved from the driver's region: 61 currencies across 83 countries.
- Debugged a recursive row-level-security policy that silently broke every chat read, and a distance algorithm that let an imprecise fix pay a parked driver. Moved feed photos from 479 KB base64 blobs in Postgres — 9.4 MB per feed — to object storage with ~20 KB thumbnails.

SELECTED ML PROJECTS

- ML Serving Platform** — versioned rollout, A/B routing, circuit breaking, per-feature drift detection and caching around inference; Triton, MLflow, Optuna, Prometheus, Redis, Kubernetes.
- Edge AI Model Compression** — distillation from a ResNet50 teacher, magnitude pruning and int8 quantization to ONNX for TSMC 3nm constraints; targeted 8× smaller, measured -6.3% and published why.
- IoT Predictive Maintenance on GCP** — LSTM failure prediction on a 48-hour horizon, split chronologically so no future reading leaks into training; BigQuery, Vertex AI, Dataflow, Pub/Sub.
- DETEK, Data-Leak Detection on AWS** — BERT content classifier paired with an LSTM behavioural model, because either alone produces unusable alerts; CloudTrail and VPC Flow collectors on Terraform infrastructure.

SKILLS

Languages: Python, TypeScript, Java, SQL, R, STATA

ML & AI: Production pipelines, training and evaluation, statistical modeling, deep learning in PyTorch and TensorFlow (LSTM, BERT, DDPM), knowledge distillation, pruning, int8 quantization, ONNX, anomaly detection, LLM-assisted diagnosis, Scikit-learn, Pandas, NumPy

Systems & Cloud: System design, distributed systems, CI/CD, GCP (BigQuery, Vertex AI, Dataflow, Pub/Sub), AWS (Lambda, S3), FastAPI, Flask, Node.js, React, Capacitor, Electron, Postgres + RLS, Redis, Docker, Kubernetes, Terraform

EDUCATION & CERTIFICATIONS

Institute of Engineering and Management, Kolkata · B.Tech, Computer Science & Engineering

2022 – 2026

Certifications: Meta Data Analyst; Google Data Analytics; Google Cybersecurity; IBM Cybersecurity Essentials; Google Cloud and AWS engineering paths; Microsoft Azure AI Cloud Apps